



NATIONAL UNIVERSITY OF MODERN LANGUAGES

Computer Networks Assignment No - 2

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Part 1: Theoretical Questions

1. Checksum

a. Concept of Checksum Technique

A checksum is an error detection technique used in data communication to ensure that data received is the same as the data sent. Before sending data, the sender adds all data words using binary addition and takes the 1's complement of the result. This final value is called the checksum. The checksum is sent along with the data.

b. How Checksum Detects Errors

When the receiver gets the data, it performs the same binary addition on the received data words, including the checksum. Then it takes the 1's complement of the total.

- If the result is **all zeros**, it means no error is detected.
- If the result contains **any 1**, it means some bits were changed during transmission and an error is detected.

c. Limitations of the Checksum Method

- It cannot detect all types of errors, especially if the error cancels itself out.
- It is weaker than CRC because it only performs addition, not polynomial division.
- Burst errors may go undetected.
- It provides no correction, only detection.

2. Selective Reject (Selective Repeat) ARQ

a. How Selective Reject Works

Selective Reject ARQ retransmits **only the specific frame** that was lost or damaged. The receiver stores correctly received frames in a buffer even if they arrive out of order. It then requests only the missing frame again.

b. Difference from Go-Back-N ARQ

- In Go-Back-N, if one frame is lost, **all frames from that point onward** are retransmitted.
- In Selective Reject, **only the missing frame** is retransmitted.
- Selective Reject requires **more memory** because the receiver must store out-of-order frames.

c. Advantages and Disadvantages

Advantages:

- More efficient; fewer frames are retransmitted.
- Better performance in networks with high error rates.

Disadvantages:

- More complex to implement.

- Requires larger buffers at the receiver.
- Out-of-order delivery must be handled.

3. Stop-and-Wait ARQ

a. Explanation with Example

In Stop-and-Wait ARQ, the sender sends **one frame at a time** and waits for an acknowledgment (ACK).

Example:

1. Sender sends Frame 1.
2. Receiver receives Frame 1 and sends ACK1.
3. After receiving ACK1, the sender sends Frame 2.

This ensures reliability but is slow.

b. Why It Is Inefficient for Long Distance

Stop-and-Wait wastes time because the sender must wait for the ACK before sending the next frame.

If distances are large, propagation delay is high, causing delays between frames and reducing throughput.

4. Go-Back-N (GBN) Protocol

a. How GBN Improves Efficiency Compared to Stop-and-Wait

GBN uses a **window** of multiple frames (for example, 4 frames) that can be sent before waiting for an ACK. This increases channel utilization and speeds up data transfer.

b. Role of Sender's Window Size

The sender window size determines how many unacknowledged frames can be in transit at a time.

A larger window size increases efficiency and improves throughput.

Part 2: Conceptual Questions

1. What Happens if an ACK Is Lost in Stop-and-Wait?

If ACK is lost, the sender waits until the timer expires. When the timer times out, the sender retransmits the same frame. The receiver, seeing the same frame again, discards it and sends the ACK again. This prevents errors from affecting communication.

2. Selective Repeat vs Go-Back-N (Efficiency and Complexity)

- Selective Repeat is more efficient because only the lost frame is resent.
- Go-Back-N is less efficient because it resends all frames after the lost one.
- Go-Back-N is simpler, while Selective Repeat is more complex and requires more memory.

3. Need for Error Detection (Checksum) Before ARQ

ARQ only handles retransmission but does not detect errors. Checksum detects if data is corrupted so the receiver knows whether to request retransmission. Without error detection, ARQ cannot function.

Part 2: Numerical Questions

1. Checksum Calculation

Given 8-bit data blocks:

10110101

11001011

01101100

Step 1: Add the three blocks using binary addition

1. 10110101

- 11001011
= 011111001 (carry out)

Take 8-bit + carry:

11111001 (drop extra carry)

2. 11111001

- 01101100
= 1 01100101 (carry out)

After adding carry:

$01100101 + 1 = 01100110$

Step 2: Take 1's complement

Checksum = 10011001

Verification:

Received data:

10110101 11001011 01101100 11101000

Add all blocks including checksum:

$11101000 + 01100110 = 101101110$

Taking final 1's complement $\neq$ 00000000

Therefore, **error is detected**.

The received data is **not error-free**.

2. Go-Back-N ARQ Calculation

Window size = 4

Total frames = 8

Assume frames 0,1,2 sent successfully.

Frame 3 is lost.

Receiver action:

Receiver expects Frame 3. It discards Frames 4, 5, 6, 7 if they arrive.

Sender action:

Sender waits for ACK for Frame 3.

Timeout occurs → sender retransmits **Frame 3 and all frames after it**.

Frames retransmitted:

Frames 3, 4, 5, 6, 7

Total retransmitted = 5 frames.

3. Stop-and-Wait ARQ Efficiency

Propagation delay = 10 ms

Transmission time = 5 ms

ACK time = 10 ms

Total cycle time = Transmission + Propagation + ACK + Propagation

Total = 5 + 10 + 10 + 10 = 35 ms

Efficiency = (Transmission time / Total cycle time)

Efficiency = 5 / 35

Efficiency = 0.142 = 14.2%